CE 459: VLSI Algorithmics

Tutorial 0. Setting up Mentor Catapult

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In this tutorial we will inspect the following topics:

- (1) How to access the Wilkinson server via ssh .
- (2) How to setup the environment variables required to run Catapult.
- (3) How to setup and license Catapult tool.
- (4) How to run Catapult for the first time.

Accessing the Wilkinson server via ssh

The first thing we need to figure out before starting using Catapult is how to access the server where these tools are installed. Several cad tools can be found in Northwestern servers, and are available for students to use with their credentials (NetID). This is the case of Mentor Graphics, which is installed and configured in [all] the Wilkinson server machines. The Wilkinson server is …

> *"..located in Tech M338, it has 22 dual-boot computers, a workgroup printer, a mobile projector, and lounge area with seating and tables. It also has 3 large TV monitors on the walls, and hardware to allow users to duplicate their laptop displays wirelessly."* - [1]

Let's take a look at the addresses of these machines (as this will come in handy later). The complete list is shown in **Table 1**:

alfred.eecs.northwestern.edu	hush.eecs.northwestern.edu
bane.eecs.northwestern.edu	joker.eecs.northwestern.edu
batgirl.eecs.northwestern.edu	killercroc.eecs.northwestern.edu
batman.eecs.northwestern.edu	madhatter.eecs.northwestern.edu
clayface.eecs.northwestern.edu	nightwing.eecs.northwestern.edu
cobblepott.eecs.northwestern.edu	poisonivy.eecs.northwestern.edu
freeze.eecs.northwestern.edu	ras.eecs.northwestern.edu
gordon.eecs.northwestern.edu	riddler.eecs.northwestern.edu
gotham.eecs.northwestern.edu	robin.eecs.northwestern.edu
harley.eecs.northwestern.edu	scarecrow.eecs.northwestern.edu
huntress.eecs.northwestern.edu	selina.eecs.northwestern.edu

Table 1. Wilkinson server machines names.

As a student, you can access any of these machines with your netid. By doing so, a folder with your netid will be created in the home directory. You can keep any project files (or any other types of files) in there, but **NOTICE THAT THESE FOLDERS ARE WIPED OUT EVERY SUMMER, SO MAKE SURE YOU BACK THEM UP ACCORDINGLY, SPECIALLY ONCE THE QUARTER FINISHES.**

There are a few options to access any of these servers **(YOU WILL NEED TO BE CONNECTED TO NORTHWESTERN VPN FOR ANY OF THESE)**. [2]

1. **Configure ssh forwarding in your local machine**.

    a. **OSX:** On Darwin (MACOS) machines, you'll need to have XQuartz installed. XQuartz is a terminal that incorporates X11 Forwarding, which is basically the package that allows graphical interfaces (like the GUI of a program you are running) to be exported. If you try to call a GUI outside of XQuartz, you will most likely get a display error, as the graphical interface of the program you called could not be exported. XQuartz documentation can be found in [3].

    Once XQuartz has been installed, you can access the server via ssh by opening XQuartz terminal and running:

    ```
    >> ssh -X [username]@[server].ece.northwestern.edu
    ```
    (1)

    where [username] stands for your netid and [server] represents any name in the list shown in **Table 1**.

    b. **Linux**: Most linux (UNIX) systems have X11 already installed so that graphical interfaces can be exported directly out of the box. In this case, all you need to do is open a Terminal Session and run **(1)**.

2. **Access machine using web browser and FastX.**

    a. Access **https://[server].ece.northwestern.edu:3300/** using any web browser, where **[server]** represents any name in the list shown in **Table 1.**

    b. Log in with your ECE credentials
    c. Click Launch Session
    d. Click on Gnome and then click Launch

Once the access to the Wilkinson server has been setup we are ready to proceed to the next step. Further information about the Wilkinson Lab can be found in [1].

Setting up Catapult Environment variables

The first thing that we have to do whenever we want to call any cad program installed in the Wilkinson servers is to set up the PATH of the libraries and tools properly. If you try to run any cad tool command line before this step, you will face a “command not found” error, because the system does not create, by default, the aliases required to run commands such as “catapult”. See **Figure 1**.

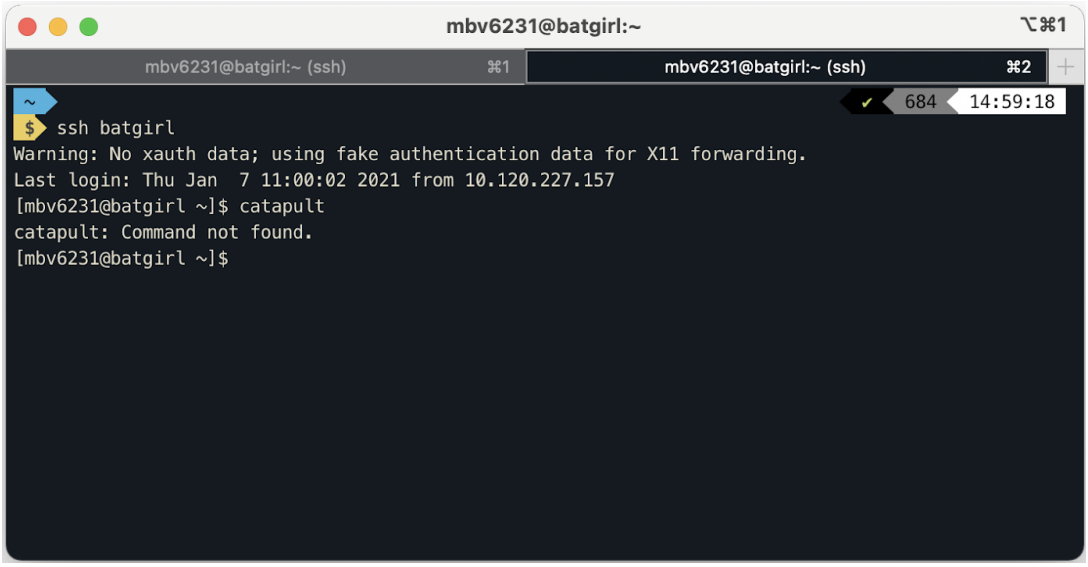

Figure 1. Command not found error.

The system, in the situation shown in **Figure 1**, does not know what the command “catapult” means, as at that specific point it has not been defined. We could create these aliases (a.k.a. definitions/links) manually every time we log into the machine, which would obviously be an unnecessary effort. Instead of that, we simply created a bash executable file (source file) which we can use to automate the creation of these aliases and definitions. This file should already be in your home directory the first time you log-in any wilkinson machine, under the name **[envsetup-catapult.sh]**.

We are not gonna get into the explicit details of how the content of this file, but just keep in mind that it basically sets up the paths to access all the cad tools that we are going to use, and that **IT MUST BE EXECUTED EVERY TIME THAT YOU ACCESS THE MACHINE.**

Thus, to set up the environment just run **(2)** before calling catapult:

```
>> source ~/envsetup-catapult.sh
```

(2)

Setting up Catapult tool

Once the environment variables have been sourced, right before we can start using Catapult, we must setup the tool itself. This step is required ONLY ONCE, the first time we are running catapult, and should not be repeated once done.

1. Access the wilkinson server and run **(3)**

```
>> catapult -product setup
```

(3)

2. A window will appear with the information shown in **Figure 2**. Choose the option "*Configure installation for your computer*" and then press "*Next*".

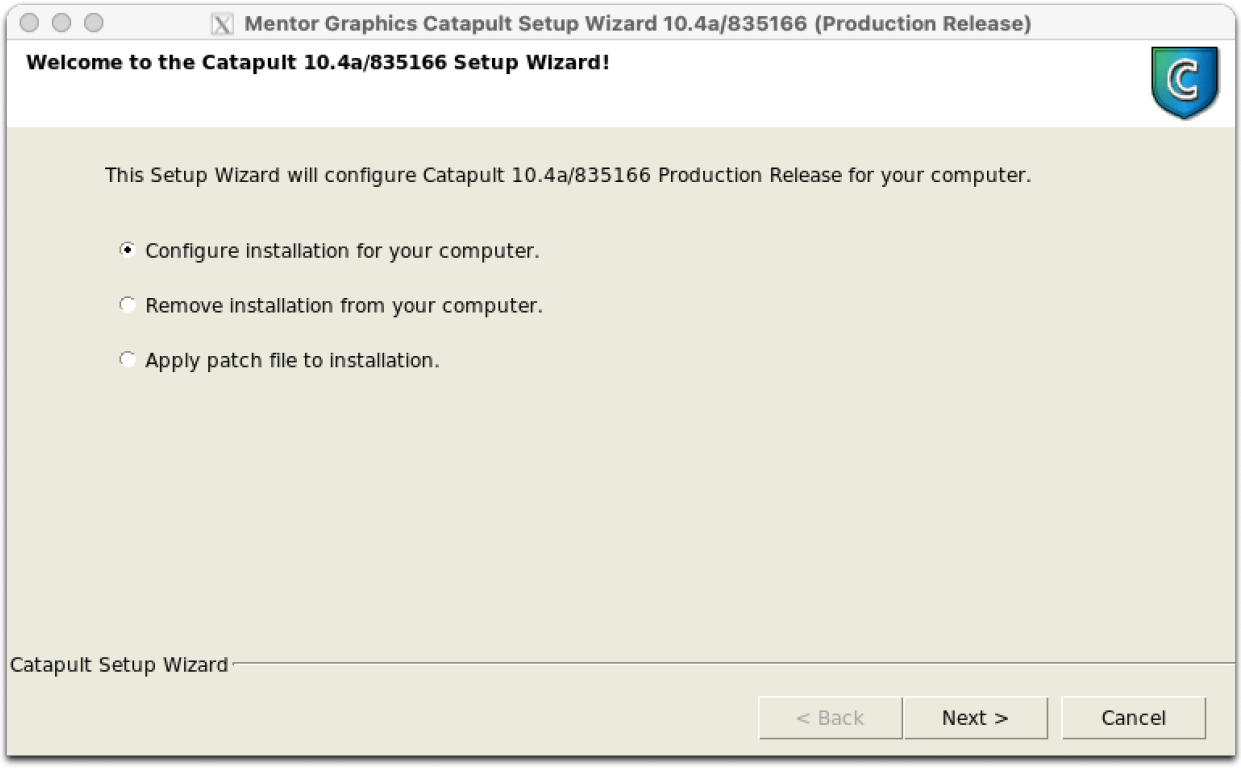

Figure 2. Catapult setup wizard initial screen.

3. Make sure that the license server information that appears in the following screen matches the one shown in **Figure 3** and press "*Verify*".

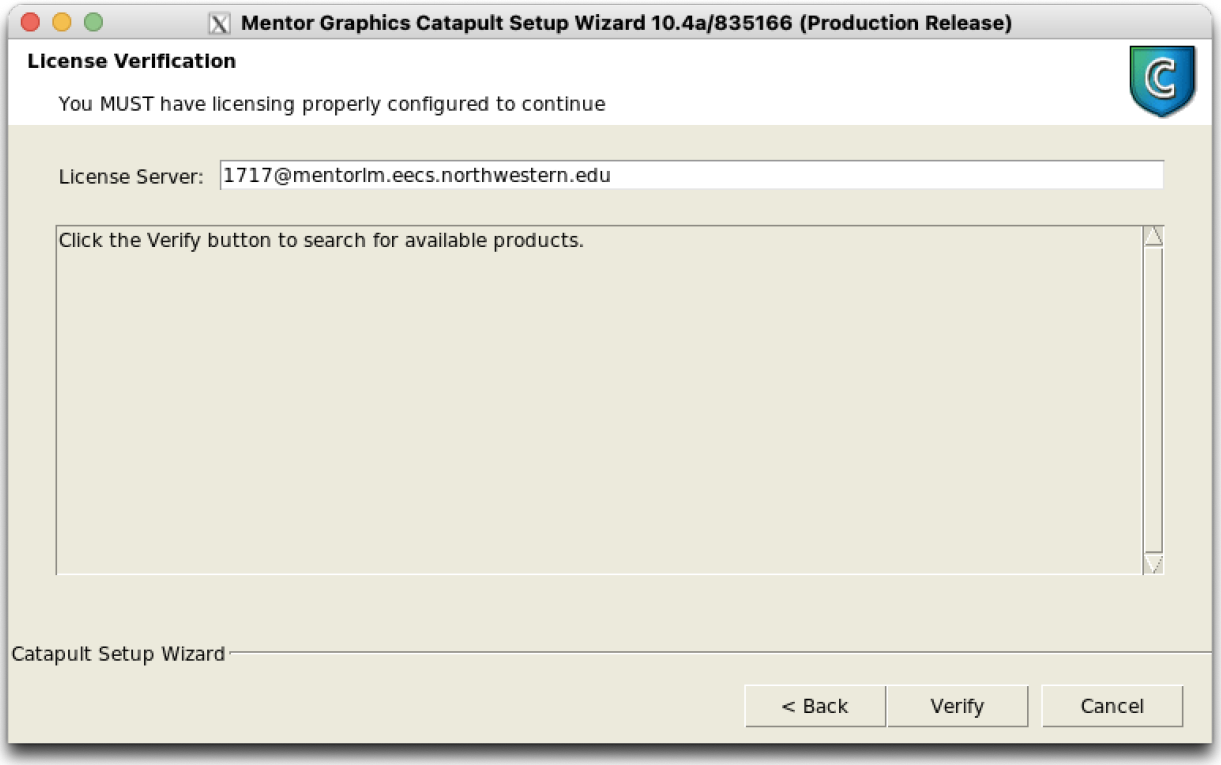

Figure 3. Catapult setup wizard license server config screen.

4. After pressing "*Verify*" make sure that the products shown in **Figure 4** appear in the screen as available. Then press "*Next*".

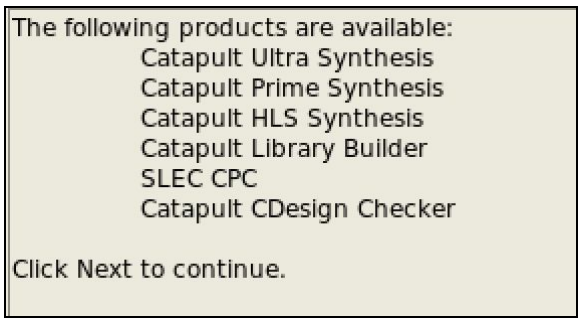

Figure 4. Catapult setup wizard available products window.

5. Make sure to select all available options in the following screen, as shown in **Figure 5**. Then press "*Next*".

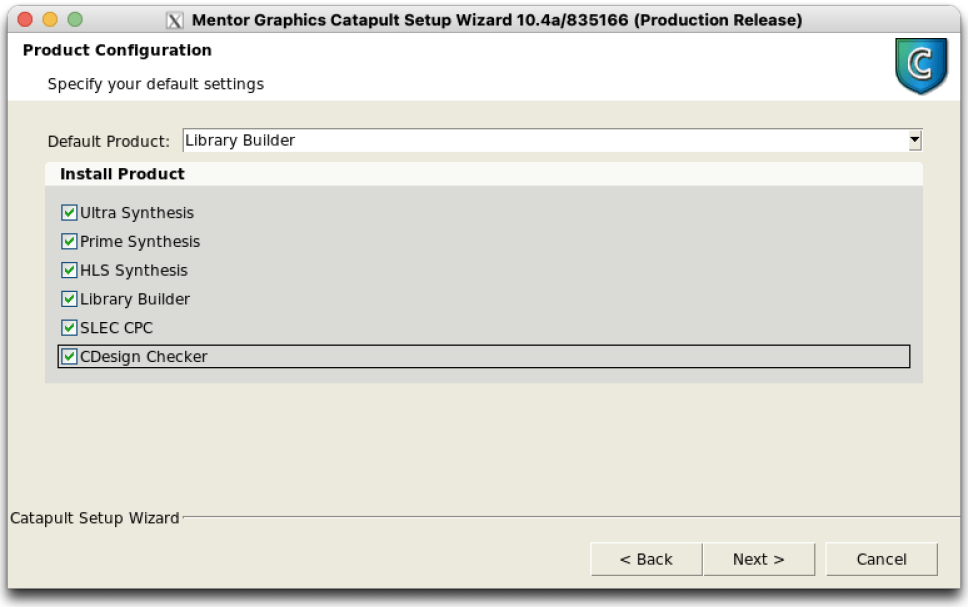

Figure 5. Catapult setup wizard available products window.

6. Read and accept the terms in the following window, then press "*Next*". Finally, press "*Install*". If the installation was successful, a screen like the one shown in **Figure 7** should appear. If so, press "*Finish*" to complete the configuration.

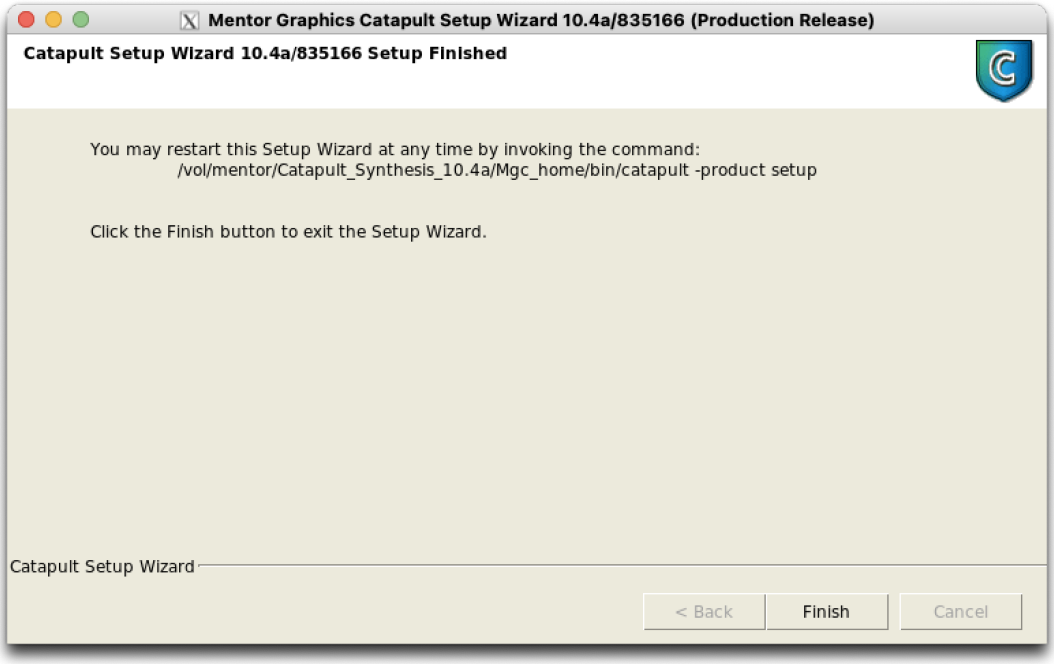

Figure 7. Catapult setup wizard successful installation screen.

Running Catapult for the first time

Once the previous steps have been completed, we can run catapult for the first time by simply invoking it using **(4)** .

```
>> catapult -product ultra
```
(4)

If you receive an error message that says "*catapult: Command not found.*" you likely did not call the environment setup (Refer to Section 2 of this tutorial). Remember that the file [**envsetup-catapult.sh**] must be called EVERY TIME YOU OPEN A NEW TERMINAL SESSION.

If you receive an error message that says:

> "*# Error: License request failed, pid = catapultcoreuv, mgls_errno = 522*
> *# Error: License server does not support this feature (catapultcoreuv)*
> *failed to obtain license*"

you likely did not setup the license server required to run mentor tools properly (Refer to Section 3 of this tutorial). Remember that this license server setup MUST BE RUN ONE SINGLE TIME AND **DOES NOT** NEED TO BE CALLED FOR EVERY TERMINAL SESSION YOU USE.

If everything went as expected, Catapult should open in a window, as shown in **Figure 8** (notice that, in case you are using ssh and exporting the window, it might take a while to open, specially if you are using a slow internet connection or if you are located far away from Evanston).

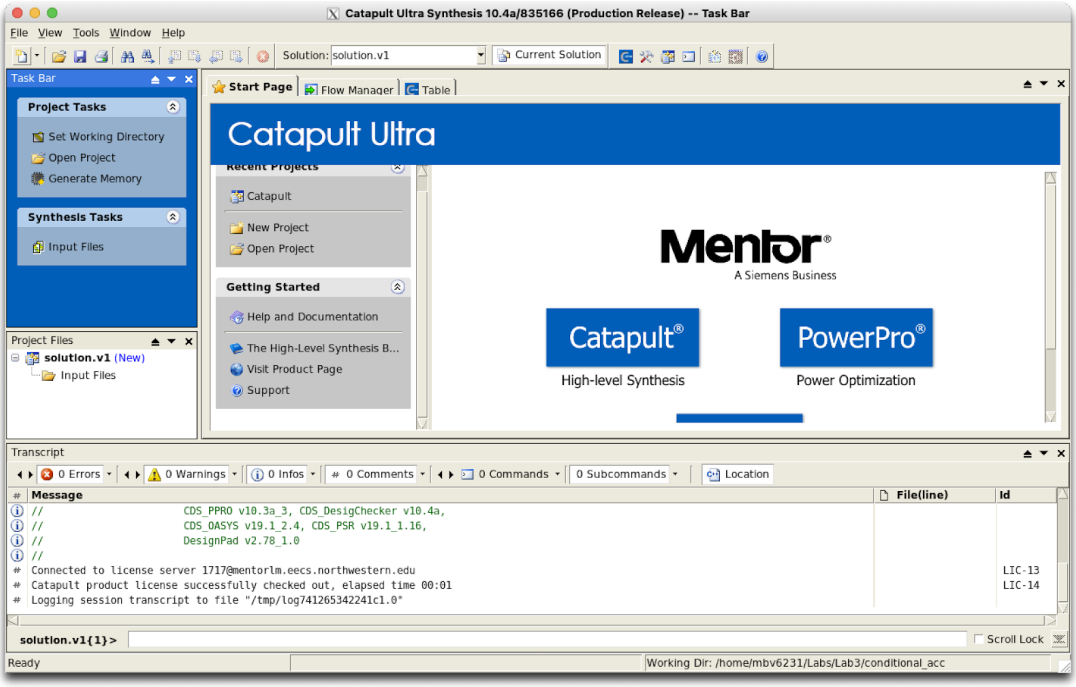

Figure 8. Catapult Ultra main screen.

References

[1] Wilkinson Lab information page:
http://it.eecs.northwestern.edu/info/2015/11/03/info-labs.html
[2] Northwestern VPN: https://www.it.northwestern.edu/oncampus/vpn/
[3] XQuartz documentation: https://www.xquartz.org/